

Python Versions

Every programming language evolves as new ideas and technologies emerge, and the developers of Python have continually made the language more versatile and powerful. As of this writing, the latest version is Python 3.11, but everything in this book should run on Python 3.9 or later. In this section, we'll find out if Python is already installed on your system and whether you need to install a newer version. Appendix A contains additional details about installing the latest version of Python on each major operating system as well.

Running Snippets of Python Code

You can run Python's interpreter in a terminal window, allowing you to try bits of Python code without having to save and run an entire program.

Throughout this book, you'll see code snippets that look like this:

```
>>> print("Hello Python interpreter!")
Hello Python interpreter!
```

The three angle brackets (`>>>`) prompt, which we'll refer to as a *Python prompt*, indicates that you should be using the terminal window. The bold text is the code you should type in and then execute by pressing ENTER. Most of the examples in this book are small, self-contained programs that you'll run from your text editor rather than the terminal, because you'll write most of your code in the text editor. But sometimes, basic concepts will be shown in a series of snippets run through a Python terminal session to demonstrate particular concepts more efficiently. When you see three angle brackets in a code listing, you're looking at code and output from a terminal session. We'll try coding in the interpreter on your system in a moment.

We'll also use a text editor to create a simple program called *Hello World!* that has become a staple of learning to program. There's a long-held tradition in the programming world that printing the message `Hello world!` to the screen as your first program in a new language will bring you good luck. Such a simple program serves a very real purpose. If it runs correctly on your system, then any Python program you write should work as well.

About the VS Code Editor

VS Code is a powerful, professional-quality text editor that's free and beginner-friendly. VS Code is great for both simple and complex projects, so if you become comfortable using it while learning Python, you can continue using it as you progress to larger and more complicated projects. VS Code can be installed on all modern operating systems, and it supports most programming languages, including Python.

Appendix B provides information on other text editors. If you're curious about the other options, you might want to skim that appendix at this